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RESEARCH NOTE

Clockwise rotation of the Korean Peninsula with respect to the North China Block inferred from an improved Early Triassic palaeomagnetic pole for the Ryeongnam Block

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SUMMARY

In order to detect a possible clockwise rotation of the Korean Peninsula with respect to the North China Block, Early Triassic rocks in the Ryeongnam Block, Korean Peninsula have been restudied. Sandstones in the Nogam Formation were collected at four sites for palaeomagnetic study. A high-temperature magnetization component with an unblocking temperature of 690 °C is isolated from all sites and yields a positive fold test at the 95 per cent confidence level. This high-temperature component is interpreted to be of primary origin because folding is Middle Triassic in age. The primary directions together with previously reported ones give a new Early Triassic mean direction ($D=347.4^\circ$, $I=23.8^\circ$, $\alpha_{95}=5.1^\circ$) and corresponding palaeomagnetic pole (62.6°N, 336.1°E, $A_{95}=4.2^\circ$) for the Ryeongnam Block. This improved Early Triassic pole for the Ryeongnam Block is located to the west of the coeval poles for the North China Block. It is therefore concluded that the Ryeongnam Block underwent clockwise rotation of $10^\circ-15^\circ$ with respect to the North China Block. Because the amount of rotation observed in the present study is comparable with rotations observed in Cretaceous results from the Ryeongnam Block and in the Triassic–Cretaceous results from other blocks in the Korean Peninsula, the whole of the Korean Peninsula appears to have been subjected to clockwise rotation in Tertiary times.

Key words: Korea, palaeomagnetism, rotation, Triassic.

1 INTRODUCTION

A relative tectonic movement between the Korean Peninsula and North China Block has been suggested by several authors (Lee *et al.* 1987; Ma *et al.* 1993; Uchimura *et al.* 1996; Zhao *et al.* 1999), who have claimed that the Korean Peninsula, as well as an area east of the Tan-Lu Fault, underwent clockwise tectonic rotation with respect to the interior part of the North China Block. The amount of clockwise rotation is estimated to be between 9° and 30°, and is considered to have occurred between Cretaceous and Tertiary times (Ma *et al.* 1993; Uchimura *et al.* 1996; Zhao *et al.* 1999).

The palaeomagnetic data supporting clockwise rotation are of Cretaceous age (Otofuji *et al.* 1986; Lee *et al.* 1987; Zhao *et al.* 1999). Although the clockwise rotation is also expected to be observed in palaeomagnetic data of older ages, an Early Triassic palaeomagnetic direction from the Korean Peninsula shows insignificant clockwise rotation with respect to the contemporaneous direction for the North China Block ($9.6^\circ \pm 10.0^\circ$, Uno 1999). More sampling is therefore required to improve the pole position of the Korean Peninsula in the Early Triassic and test for possible rotation.

A further important factor concerning the comparison of Early Triassic data for the Korean Peninsula with those for the North China Block is the choice of North China Block poles. In Uno (1999), the Early Triassic pole for the Korean Peninsula was compared with that for the North China Block compiled by Yang *et al.* (1991). The Early Triassic pole of Yang *et al.* (1991) is located closer to the pole for the Korean Peninsula than any other Early Triassic pole from the North China Block, and a clockwise rotation between the two areas is not quite demonstrably significant ($9.6^\circ \pm 10.0^\circ$). Comparisons using other data sets from the North China Block therefore need to be undertaken.

To evaluate the Cretaceous–Tertiary clockwise rotation, samples have been collected from the Lower Triassic Nogam

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Formation in the Bagjisan Syncline within the Ryeongnam Block. The Ryeongnam Block, the southeastern part of the Korean Peninsula, is a suitable tectonic unit within the Korean Peninsula with which to make a comparison with the North China Block, because the Ryeongnam Block is an original part of the North China Block (Fig. 1) and because it has provided all of the previously reported Cretaceous palaeomagnetic results. The Bagjisan Syncline is an area where a fold test is available.

2 GEOLOGICAL SETTING AND SAMPLING

The Korean Peninsula is situated to the east of the major tectonic blocks comprising East Asia (North and South China blocks; see Fig. 1a). It is composed of three tectonic divisions with basements of Archaean to Mesoproterozoic age, namely the Nangrim, Gyeonggi, and Ryeongnam blocks from north to south (Reedman & Um 1975; Cluzel *et al.* 1990) (Fig. 1b). The former two blocks are bounded by the Imjingang Belt, and the latter two by the Ogcheon Belt.

The Late Carboniferous–Early Triassic Pyeongan Supergroup is the sedimentary cover of the northeastern part of the Ryeongnam Block (Cluzel 1991; Cluzel & Cadet 1992; Otoh & Sasaki 1998). The Pyeongan Supergroup unconformably overlies the Early Cambrian–Middle Silurian Joseon Supergroup that yields Cambro-Ordovician faunas with North China affinity (Kobayashi 1966; Kobayashi 1967), and is unconformably overlain by the Late Triassic–Early Jurassic Daedong Supergroup.

The Pyeongan Supergroup is composed of four formations: the Hongjeom, Sadong, Gobangsan and Nogam formations in

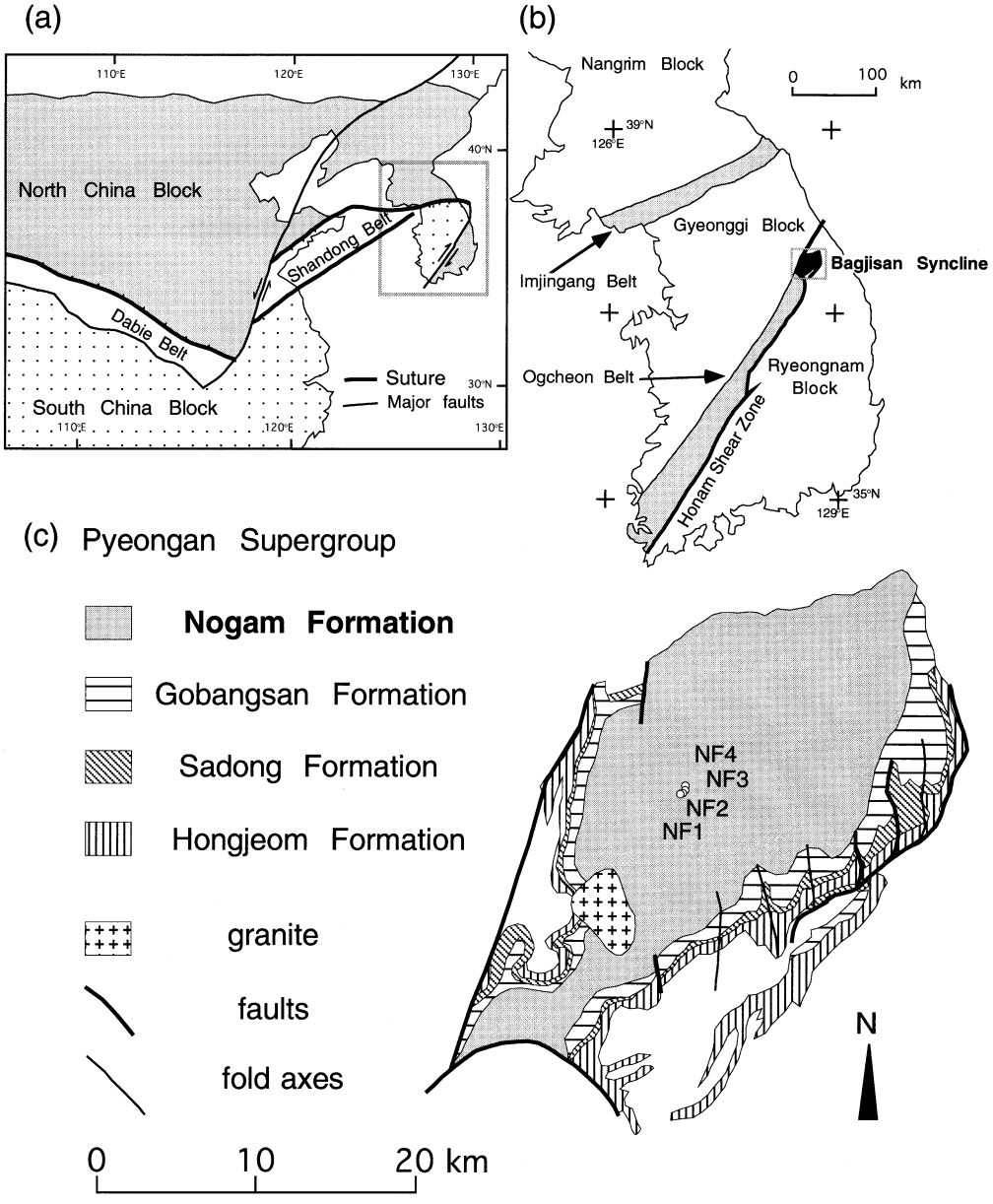


Figure 1. (a) Outline tectonic map of the Korean Peninsula and adjacent area (after Li 1994). (b) Major tectonic divisions of the Korean Peninsula. The Bagjisan Syncline (study area) is situated in the northeastern part of the Ryeongnam Block. (c) Geological map of the Bagjisan Syncline. Open circles denote the palaeomagnetic sampling sites of the Nogam Formation.

ascending order (Reedman & Um 1975; Lee 1987). The Nogam Formation, the top of the supergroup, is assigned an Early Triassic age on the basis of stratigraphic and structural data (Reedman & Um 1975; Kobayashi 1977). It consists of green, red and grey sandstones, grey shales, and conglomerates with a total thickness of 3000 m. The upper part of the Nogam Formation contains silicic volcanoclastics (Choi *et al.* 1991). The ages of the Hongjeom, Sadong and Gobangsan formations are palaeontologically dated as Late Carboniferous, Early Permian and Late Permian, respectively (Kawasaki 1939; Cheong 1974). A paraconformity is observed between the lower and upper Sadong Formation: the Hongjeom and lower Sadong formations were deposited in shallow marine basins, while the upper Sadong, Gobangsan and Nogam formations were formed in non-marine conditions (Reedman & Um 1975).

In the Bagjisan Syncline, the Pyeongan Supergroup is well exposed (Fig. 1c). The Bagjisan synclinal structure was formed after the deposition of the Early Triassic Nogam Formation and prior to Middle–Late Triassic faulting along the Mungyeong–Jeongseon Tectonic Line (also referred to as the Gongsuweon Thrust), which sharply cuts the western part of the syncline. Hence formation of the syncline was slightly earlier than fault motion and occurred during the Middle Triassic (Cluzel *et al.* 1991).

33 hand samples of greenish sandstones of the Nogam Formation were collected at four sites in the Bagjisan Syncline for palaeomagnetic measurements (Fig. 1c). One site (NF1) shows a small-scale folded structure, while the other three sites show monoclinal structures. Eight to nine hand samples were distributed over distances ranging up to 15 m in each site and were orientated by magnetic compass. Sites NF1, 2, 3 and 4 correspond to the sites SK8, 9, 28 and 27 in Uno (1999), respectively, and were restricted to this area because almost complete remagnetization has been found elsewhere in this area (Uno 1999, 2000). The present geomagnetic field declination was calculated using the International Geomagnetic Reference Field (IAGA Division V, Working Group 8 1995).

3 PALAEOMAGNETISM

In the laboratory, cores of 25 mm diameter and 22 mm length were cored from each hand sample. Natural remanent magnetizations (NRMs) were measured with a 2-G cryogenic magnetometer. All samples were subjected to progressive thermal demagnetization treatment with a Natsuhara TDS-1 thermal demagnetizer in step of 50 °C up to 500 °C. Demagnetization steps of 10, 20 or 30 °C were used in the temperature range of 500–670 °C. Above 670 °C, steps of 5 °C were employed.

The results for each sample were plotted on orthogonal vector diagrams (Zijderveld 1967) to evaluate the demagnetization behaviour. Principal component analysis (Kirschvink 1980) was used to estimate the directions of the magnetic components (Table 1).

Progressive acquisition of saturation isothermal remanent magnetization (IRM) up to the maximum field of 2.7 T and thermal demagnetization of a three-component IRM (Lowrie 1990) were performed with a 2-G pulse magnetizer to identify ferromagnetic mineral components.

3.1 Thermal demagnetization and rock-magnetic experiments

With the exception of a component removed in the first one or two steps of treatment, thermal demagnetization of the samples reveals two magnetization components at all four sites.

The high-temperature component is well-defined and converges to the origin (Fig. 2). It shows a steady decay to

Table 1. Palaeomagnetic results of the Nogam Formation.

Site	*N*	*In situ*		Tilt-corrected				Locality		Strike	Dip
		D (°)	*I* (°)	*D* (°)	*I* (°)	*k*	α_{95} (°)	Lat. (°N)	Long. (°E)	(°)	(°)
High-temperature component											
NF1	9	359.6	15.6			25.0	10.5	37.519	128.566	148*	43*
				345.3	30.7	53.1	7.1				
NF2	5	3.0	44.4	345.6	19.6	41.3	12.1	37.520	128.568	216	36
NF3	8	19.0	42.3	353.6	29.3	50.3	7.9	37.526	128.563	210	35
NF4	8	7.3	45.0	351.5	21.2	23.1	11.8	37.526	128.568	224	33
Mean	4	6.7	37.2			26.8	18.1	37.500	128.500		
				349.0	25.2	144.1	7.7				
Tr1 Mean	7			347.4	23.8	143.5	5.1	37.500	128.500		
Low-temperature component											
NF1	8	331.9	54.9			78.3	6.3	37.519	128.566	148*	43*
				289.7	41.2	25.7	11.1				
NF2	4	346.4	60.1	327.8	28.6	259.2	5.7	37.520	128.568	216	36
NF3	6	352.9	61.7	325.0	33.9	53.7	9.2	37.526	128.563	210	35
NF4	7	348.4	56.6	334.7	27.0	64.9	7.6	37.526	128.568	224	33
Mean	4	344.4	58.6			197.8	6.5	37.500	128.500		
				320.7	33.8	21.7	20.2				

N: number of samples in the calculation for the means; *D* and *I*: declination and inclination, respectively; *k*: precision parameter; α_{95}: radius of the cone of 95 per cent confidence; Lat.: north latitude; Long.: east longitude. The Early Triassic (Tr1) mean direction is calculated by combining the present data with those of Uno (1999). *Average bedding plane.

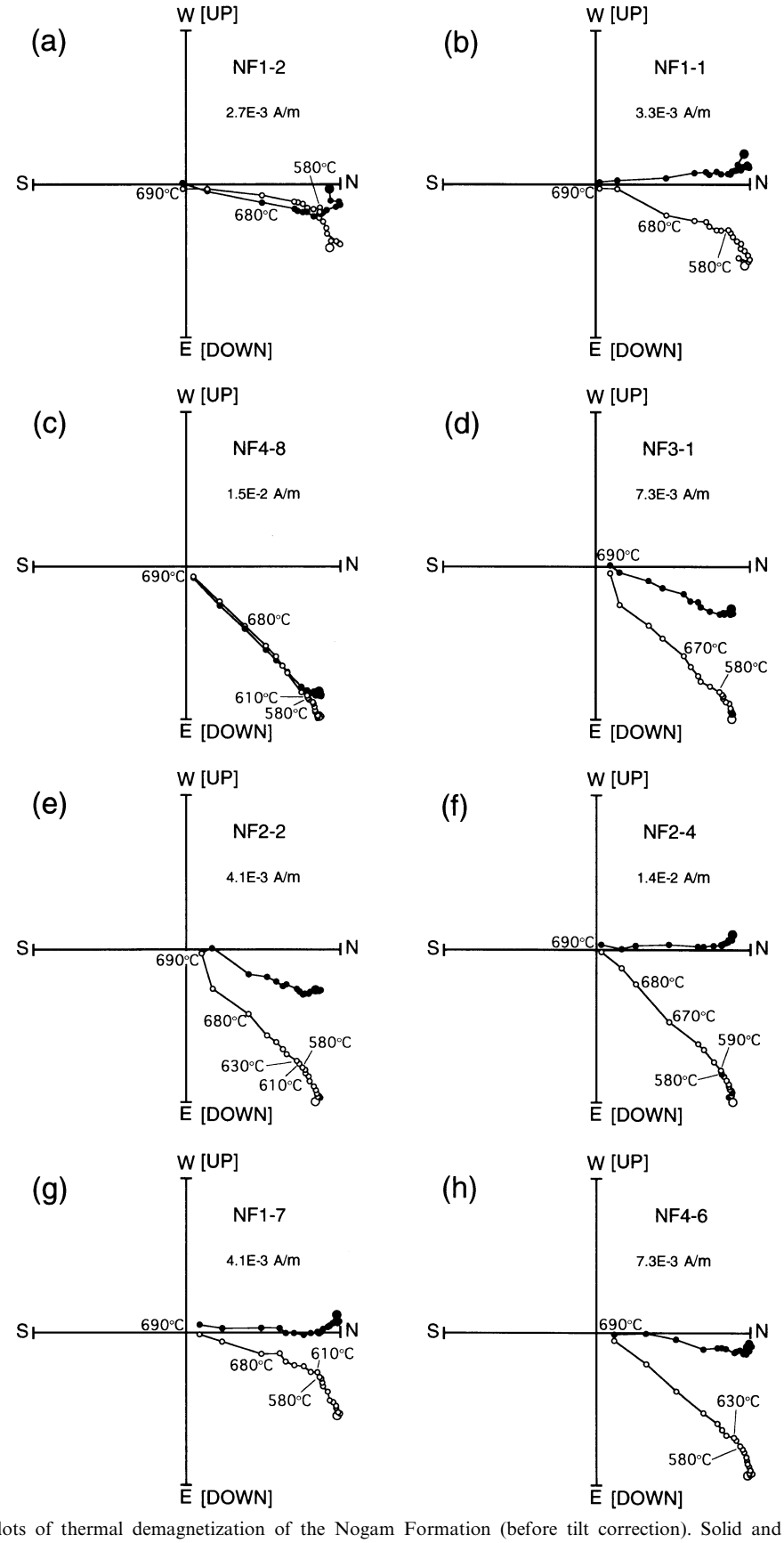


Figure 2. Orthogonal plots of thermal demagnetization of the Nogam Formation (before tilt correction). Solid and open symbols represent projection on horizontal and vertical planes, respectively.

660–670 °C, a large decay above these temperatures, and is unblocked at 690 °C, indicating a remanent magnetization carried by haematite. The low-temperature component shows a steady decay, and unblocking temperatures ranging from 580 to 630 °C (Fig. 2) with 84 per cent showing an unblocking temperature of 580 °C.

Thermal demagnetization of the composite IRM was performed on representative samples showing unblocking temperatures of 580 and 690 °C. Unblocking temperature phases at 580 and 690 °C are revealed (Fig. 3) and indicate that both magnetite and haematite are magnetic carriers.

3.2 Palaeomagnetic directions

3.2.1 High-temperature component

The directions of the high-temperature component cluster well at each site (Fig. 4a). The precision parameter of the four site-mean directions ranges between 23.1 and 53.1 (Table 1).

The four site-mean directions before tilt correction, with a mean of $D=6.7°$, $I=37.2°$ with $\alpha_{95}=18.1°$, show northerly–northeasterly declinations and variable inclinations between 15° and 45°, while these directions after tilt correction, with a mean of $D=349.0°$, $I=25.2°$ with $\alpha_{95}=7.7°$, have northwesterly declinations with shallow inclinations. Tilt correction increases the precision parameter of the mean direction from 26.8 to 144.1, and a fold test is positive at the 95 per cent confidence level (McFadden 1990), indicating that there is no grounds for rejecting the hypothesis that the high-temperature component is of pre-folding origin. Tilt correction of the single site NF1 located within a folded structure increases the precision parameter from 25.0 to 53.1. Although the latter fold test is not significant at the 95 per cent confidence level (McFadden 1990), the improvement of the clustering supports the pre-folding nature of the high-temperature component. The high-temperature component is considered to be of primary origin, because folding of the Nogam Formation occurred in the Middle Triassic as evidenced by structural data (Cluzel *et al.* 1991).

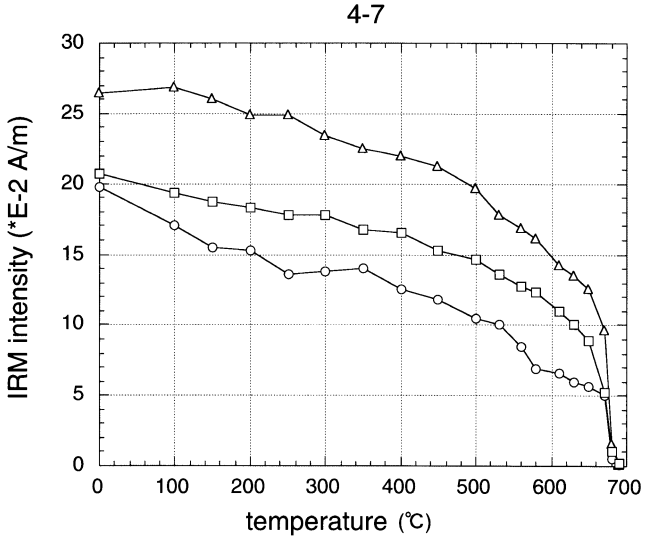


Figure 3. Thermal demagnetization of a composite IRM (Lowrie 1990): circles indicate the soft component, 0.12 T field; triangles indicate the medium component, 0.4 T field; and squares indicate the hard component, 2.7 T field.

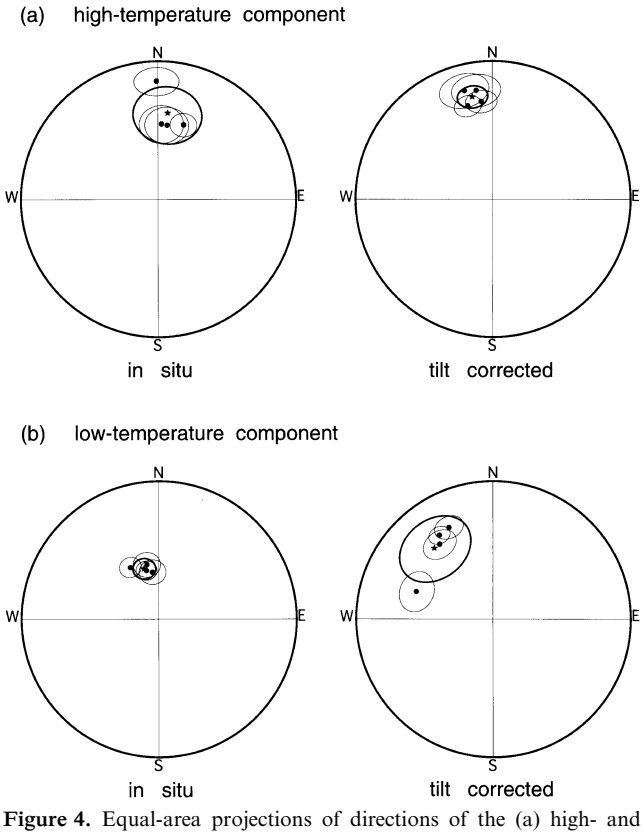


Figure 4. Equal-area projections of directions of the (a) high- and (b) low-temperature components (with associated 95 per cent confidence limits) from four sites of the Nogam Formation. Stars and bold circles represent the mean directions and associated 95 per cent confidence limits, respectively. The fold test of the high- (low-) temperature component is positive (negative) at the 95 per cent confidence level (McFadden 1990).

Since the tilt-corrected mean direction of the four sites is statistically coincident with the previously reported mean direction of seven sites from the same area ($D=347.1°$, $I=23.8°$ with $\alpha_{95}=5.5°$, Uno 1999), palaeomagnetic directions from the two studies are combined to yield a new Early Triassic mean direction and a palaeomagnetic pole for the Ryeongnam Block ($D=347.4°$, $I=23.8°$ with $\alpha_{95}=5.1°$, palaeomagnetic pole at 62.6°N, 336.1°E with $A_{95}=4.2°$).

3.2.2 Low-temperature component

The four site-mean directions of the low-temperature component have precision parameters ranging from 53.7 to 259.2 (Table 1). Mean directions before and after tilt correction are $D=344.4°$, $I=58.6°$ ($\alpha_{95}=6.5°$) and $D=320.7°$, $I=33.8°$ ($\alpha_{95}=20.2°$), respectively (Fig. 4b). The former mean direction conforms to the Earth's present-day field direction ($D=352.3°$, $I=53.4°$) and the Late Tertiary direction ($D=1.7°$, $I=50.4°$ with $\alpha_{95}=3.6°$, Lee *et al.* 1999), and a negative fold test at the 95 per cent confidence level (McFadden 1990) indicates no grounds for rejecting the hypothesis that the low-temperature component post-dates folding. The low-temperature component is therefore attributed to remagnetization in Recent or Late Tertiary times.

4 DISCUSSION

4.1 Clockwise rotation of the Ryeongnam Block with respect to the North China Block

The Early Triassic palaeomagnetic pole for the Ryeongnam Block is compared with three North China Block poles of Early Triassic age (Yang *et al.* 1991; Enkin *et al.* 1992; Embleton *et al.* 1996). Since the pole of Embleton *et al.* (1996) is based on 196 samples from six studies, a palaeomagnetic pole is recalculated using the mean directions of the six studies. The recalculated pole lies at 55.6°N, 355.0°E with $A_{95}=7.6°$ (Table 2). The poles of Yang *et al.* (1991) and Enkin *et al.* (1992) are located at 57.7°N, 352.8°E ($A_{95}=6.3°$) and 53.4°N, 358.4°E ($A_{95}=8.9°$), respectively. The Ryeongnam Block pole is significantly removed from the North China Block poles of Yang *et al.* (1991), Embleton *et al.* (1996), and Enkin *et al.* (1992) by $9.7°\pm7.6°$, $11.9°\pm8.7°$, and $14.8°\pm9.8°$, respectively (Fig. 5).

The difference in pole position between the Ryeongnam and North China blocks is ascribed to clockwise rotation of the Ryeongnam Block with respect to the North China Block, because negligible palaeolatitude contrast exists between the observed and expected palaeolatitudes at the reference point of the Ryeongnam Block (37.5°N, 128.5°E). Predicted clockwise rotations of the Ryeongnam Block pole with respect to the North China Block poles of Yang *et al.* (1991), Embleton *et al.* (1996), and Enkin *et al.* (1992) are $9.9°\pm9.7°$, $12.3°\pm11.7°$, and $14.8°\pm11.3°$, respectively. Hence it is concluded that the Ryeongnam Block has been subjected to clockwise rotation of between $9.9°\pm9.7°$ and $14.8°\pm11.3°$ relative to the North China Block since the Early Triassic. With respect to a mean pole from the three North China Block poles, the Ryeongnam Block pole is rotated $12.3°\pm7.0°$ clockwise.

4.2 Clockwise rotation of the Ryeongnam Block recorded in Cretaceous palaeomagnetic directions

The amount of clockwise rotation of the Korean Peninsula with respect to East Asia is determined to be $20.5°\pm7.8°$ based on Aptian (late Early Cretaceous) palaeomagnetic results from the Ryeongnam Block (Zhao *et al.* 1999). The amount of rotation recorded in Early Cretaceous data is comparable with that recorded in the Early Triassic data within confidence limits, and suggests that rotations observed in Early Triassic and Early Cretaceous data are the same age. It is therefore concluded that clockwise rotation of the Korean Peninsula (Ryeongnam Block) with respect to East Asia (North China Block) occurred after the Early Cretaceous.

4.3 Clockwise rotation of other blocks in the Korean Peninsula

The Nangrim and Gyeonggi blocks are situated in the northern and central parts of the present Korean Peninsula (Fig. 1). The amount of rotation for these other blocks in the Korean Peninsula with respect to the China blocks may be calculated to test whether or not the whole of the Korean Peninsula has experienced clockwise rotation. Available palaeomagnetic data from the Nangrim and Gyeonggi blocks are of Late Cretaceous and Late Triassic ages, respectively (Kim *et al.* 1993; Uchimura *et al.* 1996; Kim & Kim 1998; Uno & Chang 2000). Data from the Nangrim Block are compared with those from the North China Block, and data from the Gyeonggi Block are compared with those from the South China Block, because these blocks were originally parts of the North and South China blocks, respectively (Cluzel *et al.* 1990; Yin & Nie 1993; Li 1994).

4.3.1 The Nangrim Block

Uchimura *et al.* (1996) report a Late Cretaceous pole for the northern part of the Nangrim Block at 59.3°N, 202.6°E ($A_{95}=6.0°$), which can be compared with the Cretaceous pole for the North China Block at 78.6°N, 202.6°E ($A_{95}=6.2°$) (Gilder & Courtillot 1997). The discrepancy between these two pole positions is ascribed to clockwise rotation of the Nangrim Block through $25.7°\pm13.1°$ with respect to the North China Block. Since this rotation is recorded in Late Cretaceous data, the rotational motion occurred after the Late Cretaceous.

4.3.2 The Gyeonggi Block

A Late Triassic pole representative of the Gyeonggi Block is reported by Uno & Chang (2000), and located at 31.9°N, 220.2°E ($A_{95}=12.7°$). A comparison with the coeval pole for the South China Block (44.8°N, 223.6°E with $A_{95}=9.3°$, Opdyke *et al.* 1986) shows an inconclusive clockwise rotation of the Gyeonggi Block through $13.1°\pm20.4°$ with respect to the South China Block. Although a relatively large uncertainty in the rotation is present, the clockwise sense of motion implies a clockwise rotation of the whole Korean Peninsula.

Table 2. Early Triassic palaeomagnetic poles for the Ryeongnam, North China and South China blocks.

Locality	Age	*N* (*n*)	Pole		A_{95} (°)	Rotation (°)	Reference
			Lat. (°N)	Long. (°E)			
RB	Tr1	(7)	62.6	336.1	4.2		This study
NCB	Tr1	3	57.7	352.8	6.3	9.9±9.7	Yang *et al.* (1991)
NCB	Tr1	6	55.6	355.0	7.6	12.3±11.7	Embleton *et al.* (1996)
NCB	Tr1	5	53.4	358.4	8.9	14.8±11.3	Enkin *et al.* (1992)
NCB mean	Tr1	3	55.6	355.5	4.1	12.3±7.0	

RB: Ryeongnam Block; NCB: North China Block; Tr1: Early Triassic; N (*n*): the number of the studies (sites); Lat.: north latitude; Long.: east longitude; A_{95}: radius of the circle of 95 per cent confidence. The mean pole of Embleton *et al.* (1996) is recalculated from the poles for six studies. Rotation is evaluated by comparison between the observed and expected declinations at the reference point of the Ryeongnam Block.

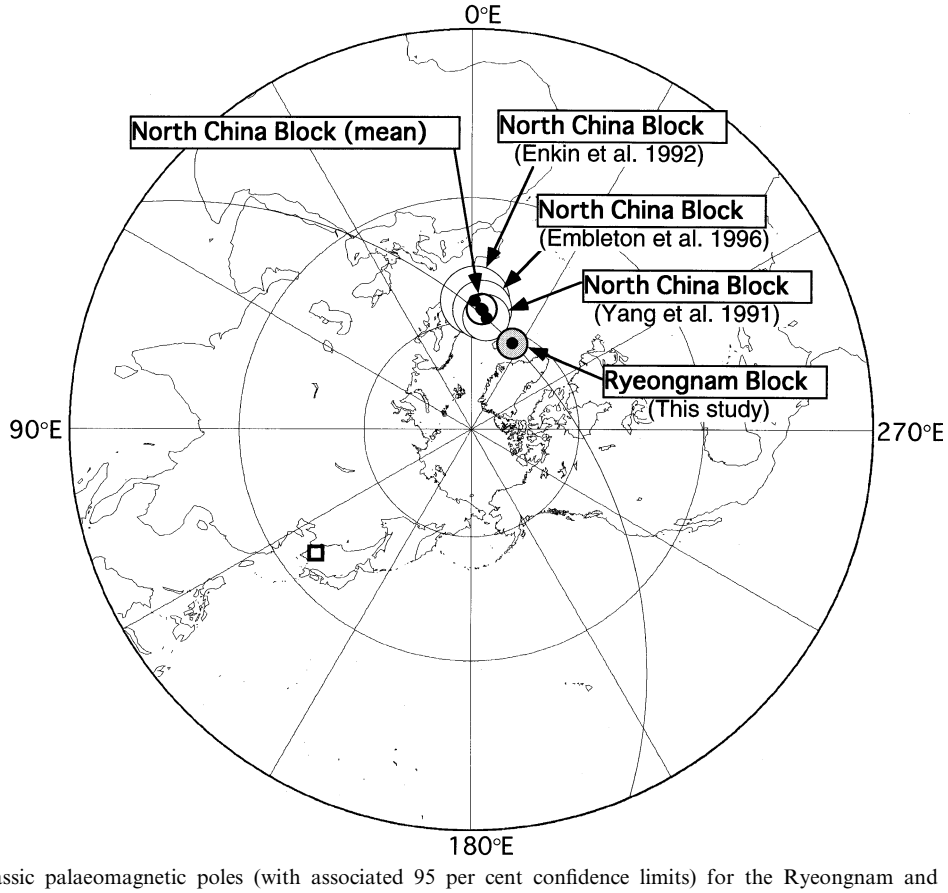


Figure 5. Early Triassic palaeomagnetic poles (with associated 95 per cent confidence limits) for the Ryeongnam and North China blocks. The open square denotes the reference point of the Ryeongnam Block, and the small circle is centred on the reference point. The Ryeongnam Block pole shows significant clockwise rotation between $9.9° \pm 9.7°$ and $14.8° \pm 11.3°$ with respect to the coeval poles for the North China Block. Data from Table 2.

4.4 Tectonic implications

The post-Cretaceous timing of clockwise rotation agrees with the timing of fault motion along the Tan-Lu Fault. Uchimura *et al.* (1996) suggested Early Tertiary faulting of the Tan-Lu Fault based on the distribution of Palaeogene basalts along the fault and concluded that sinistral motion along the fault gave rise to clockwise rotation of an area east of the Tan-Lu Fault. Because the Korean Peninsula is part of the area east of the Tan-Lu Fault, the fault motion in Early Tertiary times is considered to have caused the clockwise rotation of the whole Korean Peninsula.

5 CONCLUSIONS

An improved Early Triassic pole for the Ryeongnam Block in the southeastern part of the Korean Peninsula reveals a discordance of Early Triassic poles between the Ryeongnam and North China Blocks. The amount of clockwise rotation evaluated in this study is between $9.9° \pm 9.7°$ and $14.8° \pm 11.3°$. A consistent clockwise rotation of Triassic and Cretaceous units in three tectonic divisions (Ryeongnam Block to Nangrim Block) is observed in the Korean Peninsula. Evaluation of Mesozoic data shows that the whole of the Korean Peninsula experienced clockwise rotation in the Early Tertiary.

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